

Comprehensive Data Science Report: Melbourne Housing Market Analysis

1. Executive Summary

This report outlines a multi-phase data science project focused on cleaning, engineering, and analyzing the Melbourne Housing dataset. The project progressed from initial data sanitation to advanced feature engineering, exploratory data analysis (EDA), and finally, a mathematical assessment of data distributions and similarities.

2. Phase 1: Data Cleaning and Preparation

The primary goal was to ensure the dataset's integrity by addressing common issues in raw data.

- **Missing Values:** Handled null entries across critical columns such as BuildingArea and YearBuilt.
- **Outlier Management:** Utilized the **Interquartile Range (IQR)** and **Capping** techniques to mitigate the impact of extreme values in property prices and land sizes.
- **Data Consistency:** Removed duplicate records and standardized data types to prepare for analytical processing.

3. Phase 2: Feature Engineering

In this phase, the dataset was enriched by creating and transforming features to improve future model performance:

- **Encoding:** Applied **One-Hot Encoding** for nominal categorical variables and **Ordinal Encoding** for ranked data.
- **Scaling:** Performed feature scaling to normalize numerical ranges.
- **Feature Creation:** Developed **Domain Features** and **Interaction Features** (e.g., combining property attributes) to capture complex relationships.
- **Transformation:** Applied **Log Transform (log1p)** to reduce skewness in features and utilized **Binning** for continuous variables.

4. Phase 3: Exploratory Data Analysis (EDA)

Visual and statistical analysis was performed to uncover patterns and correlations within the Melbourne housing market:

- **Distributions:** Used **Histograms** and **Kernel Density Estimate (KDE)** plots to visualize the spread of housing prices.
- **Categorical Insights:** Employed **Grouped Boxplots** to compare price variations across different property types and regions.
- **Relationships:** Created a **Correlation Heatmap** to identify strong linear relationships between features, and **Scatter Plots** to visualize price trends relative to variables like distance from the CBD.

- **Summarization:** Generated **Groupby Summaries** to obtain a high-level view of market metrics by suburb and region.

5. Phase 4: Mathematical Analysis

The final phase focused on rigorous mathematical computations using **NumPy** to validate data characteristics:

- **Descriptive Statistics:** Computed the precise **Mean** and **Standard Deviation** for core numerical features.
- **Standardization:** Transformed data into a standard normal distribution to allow for fair comparison across features.
- **Similarity Metrics:** Calculated **Cosine Similarity** to measure the orientation/closeness of different data points within the feature space.
- **Probabilistic Estimation:** Estimated the **Probability** of specific events occurring within the dataset to better understand market risks and opportunities.

6. Conclusion

The project successfully transformed raw housing data into a refined, high-quality dataset ready for predictive modeling. By moving from systematic cleaning to advanced mathematical analysis, we have established a clear understanding of the factors driving Melbourne's real estate values.